Q1) Structure factor and intensity calculations (30 pts)

(a) Calculate the structure factor of a cubic F crystal with a single atom basis and identify its systematic absences.

Solution:

There are four lattice points per unit cell in a cubic F: (0,0,0), (1/2,1/2,0), (0,1/2,1/2), (1/2,0,1/2). These four sites are occupied by four atoms (N=4) of the same kind, with $f_i = f$.

$$S_{hkl} = \sum_{j=1}^{N} f_i e^{-2\pi i(hx+ky+lz)} = f\left(1 + e^{-\pi i(h+k)} + e^{-\pi i(k+l)} + e^{-\pi i(h+l)}\right)$$

If h, k, l are all odd, then $S_{hkl} = 4f$. Similarly, if h, k, l are all even, then $S_{hkl} = 4f$. Indices that are mixed even, odd yield $S_{hkl} = 0$.

(b) Derive the relationship for the interplanar spacing of the cubic F unit cell, with lattice parameter, *a*.

Solution:

$$\vec{k}^*_{hkl} \cdot \vec{k}^*_{hkl} = \frac{1}{d^2_{hkl}} = [h \quad k \quad l][\Gamma^*]\begin{bmatrix} h \\ k \\ l \end{bmatrix} \qquad \Gamma^*_{cubic} = \begin{bmatrix} 1/a^2 & 0 & 0 \\ 0 & 1/a^2 & 0 \\ 0 & 0 & 1/a^2 \end{bmatrix}$$

Show that $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$

(c) Using Bragg's law and assuming λ=1.54 Å and *a* = 5.64 Å, calculate the ten lowest Miller index interplanar spacings and diffraction angles.

Solution:

Bragg's law: $2d_{hkl}\sin\theta_{hkl}=\lambda$; $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}} = \frac{\lambda}{2\sin\theta_{hkl}}$;

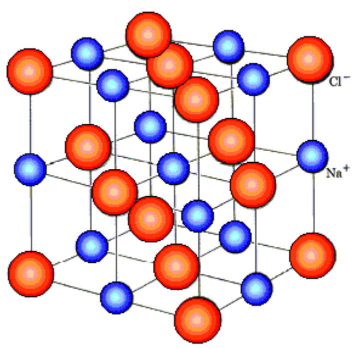
$$\theta_{hkl} = \sin^{-1}\left(\frac{\lambda}{2a}\sqrt{h^2+k^2+l^2}\right) = \sin^{-1}\left(\frac{\lambda}{2d_{hkl}}\right)$$

Prepare a table; note that diffraction angle is $2\theta_{hkl}$. ID of diffraction peaks present is optional.

Hkl	$h^2 + k^2 + l^2$	d_{hkl} (Å)	$2\theta_{hkl}$ (°)	Present?
100	1	5.64	15.69	N
110	2	3.99	22.26	N
111	3	3.256	27.36	Y
200	4	2.82	31.69	Y
210	5	2.52	35.55	N
211	6	2.30	39.07	N
220	8	1.99	45.43	Y
221	9	1.88	48.36	N
300	9	1.88	48.36	N
301	10	1.78	51.15	N

(d) Calculate the structure factor of NaCl and compute the intensity, I_{hkl}.

Solution:



The NaCl structure is composed of two cubic F sublattices.

$$\text{Na:}\quad (000)\left(\frac{1}{2}\frac{1}{2}0\right)\left(\frac{1}{2}0\frac{1}{2}\right)\left(0\frac{1}{2}\frac{1}{2}\right)$$

$$\text{Cl:}\quad \left(0\frac{1}{2}0\right)\left(\frac{1}{2}00\right)\left(00\frac{1}{2}\right)\left(\frac{1}{2}\frac{1}{2}\frac{1}{2}\right)$$

$$S_{hkl} = \sum_{j=1}^{N} f_i e^{-2\pi i(hx+ky+lz)}$$

$$= f_{Na}\left(1 + e^{-\pi i(h+k)} + e^{-\pi i(h+l)} + e^{-\pi i(k+l)}\right) + f_{Cl}\left(e^{-\pi ik} + e^{-\pi ih} + e^{-\pi il} + e^{-\pi i(h+k+l)}\right)$$

(e) Identify the systematic absences of NaCl. How do they differ from those of the cubic F lattice?

Solution:

If h, k, l are all odd, then $S_{hkl} = 4f_A - 4f_B$. When h, k, l are all even, then $S_{hkl} = 4f_A + 4f_B$. The systematic absences are the same as cubic F, but the intensity of the peaks is modulated by the even vs odd hkl indices.

In complicated cases such as this, it helps to use test indices:

hkl	S_{hkl}
100	0
110	0
111	$4f_A - 4f_B$
200	$4f_A + 4f_B$
210	0
211	0
220	$4f_A + 4f_B$

(f) Using (a) and (d), compare the intensity of the first five diffractions present in the cases of cubic F and NaCl.

Solution:

	Cubic F	NaCl
hkl	$\lvert S_{hkl}\rvert^2$	$\lvert S_{hkl}\rvert^2$
100	0	0
110	0	0
111	$16f^2$	$16(f_A - f_B)^2$
200	$16f^2$	$16(f_A + f_B)^2$
210	0	0
211	0	0
220	$16f^2$	$16(f_A + f_B)^2$
221	0	0
300	0	0
310	0	0
311	$16f^2$	$16(f_A - f_B)^2$
222	$16f^2$	$16(f_A + f_B)^2$

(g) Discuss how the intensities in (f) may be used to distinguish an elemental cubic F from a binary cubic F, such as NaCl.

Solution:

While systematic absences are identical, the difference lies in the intensity distribution of all-odd and all-even indices, with the latter (all-even indices) exhibiting systematically higher intensity (and equal to cubic F) than the (all-odd indices).

Q2) Designing an XRD experiment to detect phase transition (10 pts)
$BaTiO_3$ has a structure that is completely analogous to $SrTiO_3$, except that it is tetragonal. Assume that $SrTiO_3$ and $BaTiO_3$ form solid solutions, $(Sr,Ba)TiO_3$, but at an undetermined concentration of Ba, the solid solution takes on a tetragonal structure. Outline a powder diffraction experiment that could be used to determine the concentration of Ba at which the material transforms from cubic to tetragonal. Be sure to explain the signature that the transformation would have on the diffraction experiment.

Solution:
The cubic and tetragonal phases of the same compound will have unit cells such that a=b=c in the cubic case and a=b≠c in the tetragonal case. In a cubic case, (n00) and (00n) have the same d-spacing and same diffraction angle. However, in a tetragonal case (n00) and (00n) have different d-spacings and diffraction angles, leading to formation of new diffraction peaks.

As soon as the phase transforms to tetragonal, there is appearance of new diffraction peaks, which we can compute by considering the interplanar spacing as well as Bragg's law:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

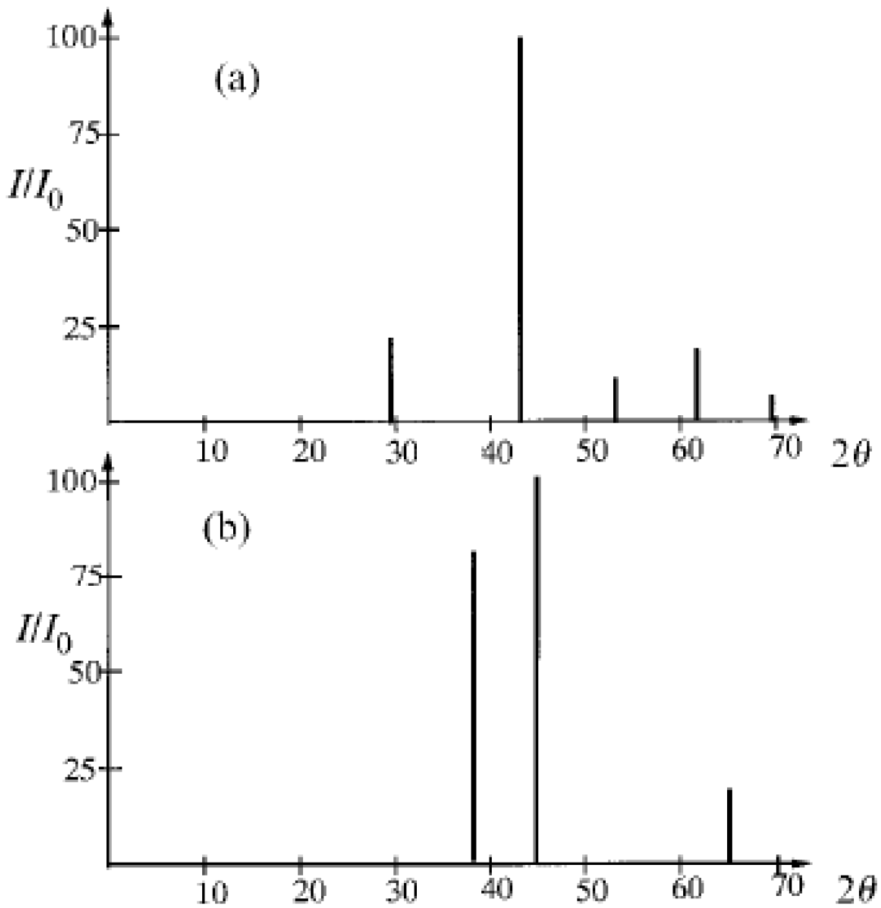
$$d_{hkl} = \frac{1}{\sqrt{\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}}}$$

$$2d_{hkl}\sin\theta_{hkl} = \lambda; \ \theta_{hkl} = \sin^{-1}\left(\frac{\lambda}{2d_{hkl}}\right)$$

Q3) Compound identification (20 pts)
Consider the reaction of Al powder and Fe powder to form FeAl. The structural data for these materials are given in the table below.

Material	space group	lattice param.	structure type
Fe	$Im\bar{3}m$	$a_0 = 2.87$ Å	bcc
Al	$Fm\bar{3}m$	$a_0 = 4.08$ Å	fcc
FeAl	$Pm\bar{3}m$	$a_0 = 3.01$ Å	B2 or CsCl

One of the schematic diffraction patterns shown below is of the combined, unreacted powders of Al and Fe, and the other is of the intermetallic phase, FeAl. Distinguish between the two. All computations and reasoning on which you base your decision must be included with your answer. Assume that the X-ray wavelength is 1.54 Å and the experimental resolution is limited to ± 0.5° in 2θ.



Solution:

Fe, Al and FeAl are cubic with I, F, and P lattices, respectively. The three compounds therefore exhibit different systematic absences.

Table 4.1. Systematic absences for centred unit cells.

Centring	Points equivalent to 0,0,0	Condition for observed intensity	Fraction of observed data
P	none	none	1
A	0, ½, ½	$k + l = 2n$	½
B	½, 0, ½	$h + l = 2n$	½
C	½, ½, 0	$h + k = 2n$	½
I	½, ½, ½	$h + k + l = 2n$	½
F	0, ½, ½ and ½, 0, ½ and ½, ½, 0	h, k, l all even or all odd	¼
R	⅓, ⅔, ⅔ and ⅔, ⅓, ⅓	$-h + k + l = 3n$	⅓

Using Bragg's law, we can compute the d_{hkl} and θ_{hkl} values for the three compounds using trial indices and noting systematic absences.

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

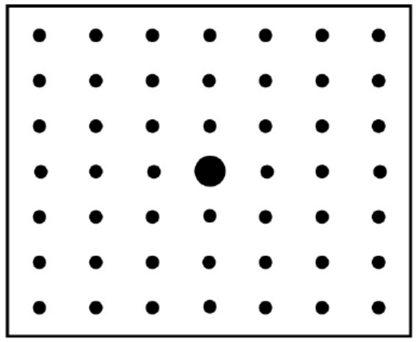
$$2d_{hkl}\sin\theta_{hkl} = \lambda;\ \theta_{hkl} = \sin^{-1}\left(\frac{\lambda}{2d_{hkl}}\right)$$

		Fe: cubic I (a=2.87A)			Al: cubic F (a=4.08A)			FeAl: cubic P (a=3.01A)		
hkl	$h^2+k^2+l^2$	d_{hkl}	2θ	Obs	d_{hkl}	2θ	Obs	d_{hkl}	2θ ?	
100	1			**N**				3.01	**29.6**	**Y**
110	2	2.03	**44.58**	**Y**				2.13	**42.4**	**Y**
111	3			**N**	2.36	**38.1**	**Y**	1.74	**52.5**	**Y**
200	4	1.44	**64.65**	**Y**	2.04	**44.4**	**Y**			**Y**
210	5			**N**						**Y**
211	6	1.17	**82.31**	**Y**						**Y**
220	8	1.01		**Y**	1.44	**64.7**	**Y**			**Y**

Based on the above table, the diffraction in (b) matches Al as all three peaks 111, 200 and 220 are a match. The first three diffractions of FeAl match those of the diffraction pattern in (a).

Q4) Diffraction pattern (30 pts)

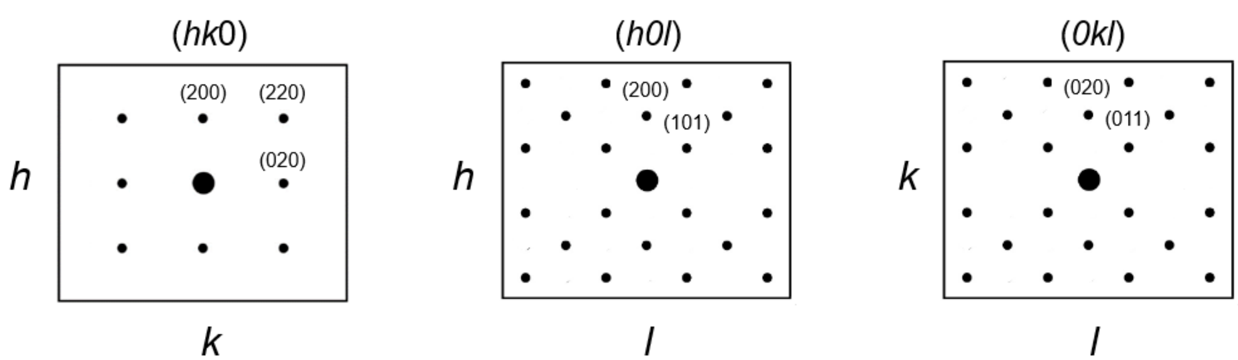
Using the reciprocal lattice pattern, draw the hk0, h0l, and 0kl sections of the diffraction patterns of the following space groups considering all systematic absences; explain your answer:



a) $I4_1/a$

I: Reflectons present when h+k+l=2n; even.
4_1: Reflections present when l=4n, involved in 00l
a: Reflections present when h=2n; even, involved in hk0

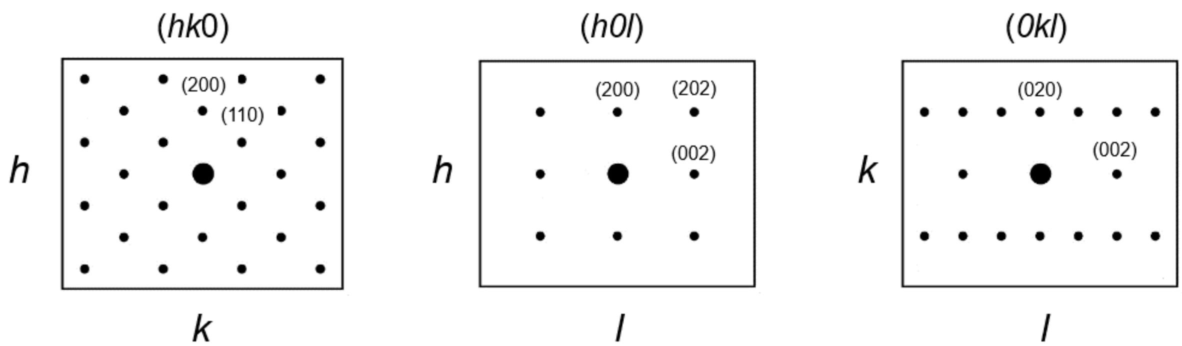
hk0: h has to be even from glide, k has to be even from I center since h is even
h0l: l=4n along 00l direction from screw. h+l has to be even aside from 00l due to I center
0kl: l=4n along 00l direction from screw. k+l has to be even aside from 00l due to I center



b) $Cmc2_1$

C: Reflections present when h+k=2n; even
c: Reflections present when l=2n; even, involved in h0l
2_1: Reflections present when h, k, or l=2n; even, involved in 00l

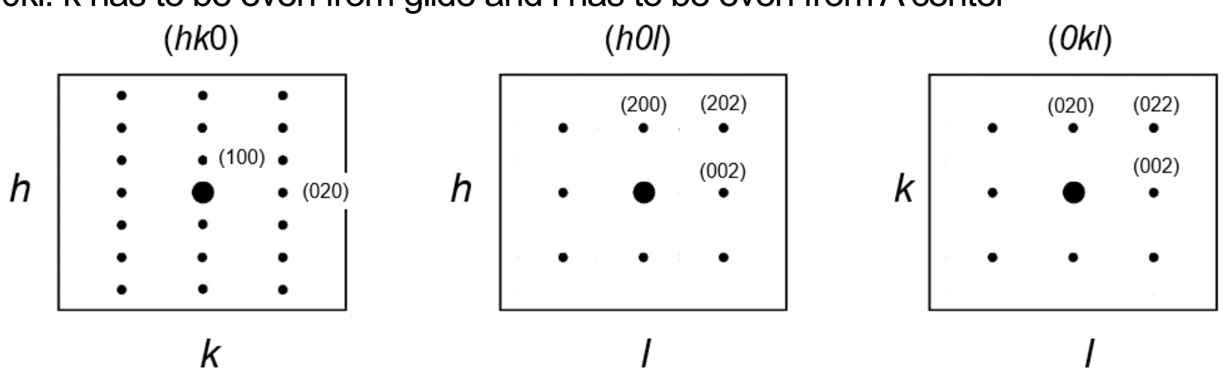
hk0: h+k has to be even from C center
h0l: l has to be even along 00l direction from screw and in general from glide, h has to be even from C center
0kl: l has to be even along 00l direction from screw, h has to be even from C center



c) Aba2

A: Reflections present when k+l=2n; even
b: Reflections present when k=2n; even, involved in 0kl
a: Reflections present when h=2n; even, involved in h0l

hk0: k has to be even from A center
h0l: h has to be even from glide and l has to be even from A center
0kl: k has to be even from glide and l has to be even from A center



Q5) Diffraction Intensity (10 pts)

List all the factors that can influence the intensity of a diffraction intensity.